

CHEM 101 Topic 2: Fundamental Measurements — Answer Key

Section 1: The Metric System & Dimensional Analysis

Problem 1: 750 mg

Problem 2: 0.500 L

Problem 3: 0.250 mg

Section 2: Scientific Notation

Problem 4: 4.5×10^{-7} m

Problem 5: 0.0032 g

Section 3: Significant Figures

Problem 6: a) 3, b) 2, c) 2

Problem 7: 6.8 g

Problem 8: 8.3 cm^3

Section 4: Temperature & Density

Problem 9: 101.3°F and 311.5 K

Problem 10: 1.08 g/mL; More dense than water.